

# Notes: Prat (JPE, 2018)

## Media Power

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# 1 Big picture

- **Question.** How can we measure the political power of a media organization when that power operates through voters' information rather than through ordinary product-market prices or quantities?
- **Core definition.** Media power is the ability of a media owner to induce voters to make electoral decisions that they would not make if reporting were unbiased.
- **Main conceptual move.** Instead of defining media concentration by platform-level market shares, the paper takes the *voter's information set* as the unit of analysis.
- **Data object.** The key input is a media-consumption matrix: for each voter segment, which news sources does the voter follow?
- **Attention overlap.** A media outlet is powerful not only when many voters follow it, but when those voters follow few other sources. A source with modest reach can be powerful if its audience gives it a large share of attention.
- **Robust-bounds approach.** The paper does not try to infer actual persuasion in every election. Instead, it derives an upper bound on how much a media owner *could* distort the vote under worst-case assumptions.
- **Theory result.** If a media group reaches no more than half of voters, a simple upper bound to its power is a function of its attention share: a group with attention share  $\alpha_G$  has power bounded by  $\min\{1, \alpha_G/(1 - \alpha_G)\}$ .
- **Empirical result.** Applying the index to the United States with 2012 Pew media-consumption data, television-centered organizations dominate. News Corporation has the largest upper-bound power index, followed by Comcast and Time Warner.
- **Calibration result.** Calibrating the model with empirical estimates of Fox News effects produces smaller absolute power levels, but the ranking remains similar.
- **Policy result.** The index can be used for merger analysis. It can produce different conclusions than simple reach-based or platform-based concentration measures.
- **Bottom line.** Media regulation should complement market-centered measures with voter-centered measures that account for cross-platform consumption and overlapping audiences.

# 2 Incremental contribution of this paper

- **From market shares to voter minds.** Existing media-concentration measures usually start from markets, platforms, firms, or revenue shares. This paper instead starts from the set of news sources that each voter consumes.
- **Cross-platform framework.** The method naturally aggregates television, newspapers, radio, websites, and online aggregators because all of them enter the same voter information environment.
- **Attention-share correction.** The paper distinguishes reach from attention. A source that reaches many voters who also consume many other outlets may have less political power than a source with a smaller but more captive audience.
- **Worst-case political power index.** The contribution is not another estimate of actual media persuasion. It is a robust upper-bound index that regulators can use when they are concerned about the maximum possible political influence of media ownership.
- **Theoretical robustness.** The upper-bound logic is designed to be robust to voter beliefs about media capture and to limits on voter attention or recall.

- **Practical implementation.** The paper translates the theory into a feasible index that can be computed from survey data on individual media consumption.
- **Merger-policy application.** The paper shows that media mergers should be evaluated using overlap in voter attention, not only using platform rules or aggregate reach.
- **New empirical ranking.** The U.S. application documents that traditional television organizations remain the most politically powerful in this metric, despite the rise of online news.

## 3 Data and measurement

### 3.1 Media-consumption matrix

The paper’s central data object is a matrix that records which news sources are followed by each voter segment. Let  $\mathcal{M}$  denote the set of all media sources and let  $M \subseteq \mathcal{M}$  denote the subset of sources followed by a given segment of voters. The share of voters in segment  $M$  is  $q_M$ , with

$$\sum_{M \subseteq \mathcal{M}} q_M = 1. \quad (1)$$

- A voter segment is defined by the exact bundle of sources the voter follows.
- A voter may follow one source or many sources.
- A media organization may own one or several sources.

**Interpretation:** This matrix captures a form of concentration that standard platform-level market shares miss: two media owners can have the same audience share but very different political power if one owner’s audience is more isolated from alternative sources.

### 3.2 Reach and attention share

The **reach** of outlet  $m$  is the share of voters who follow it:

$$Reach_m = \sum_{M: m \in M} q_M. \quad (2)$$

The **attention share** of outlet  $m$  is the weighted average of the reciprocal of the number of sources followed by each voter who follows  $m$ :

$$\alpha_m = \sum_{M: m \in M} \frac{q_M}{|M|}. \quad (3)$$

For a group of sources  $G$ , the attention share is

$$\alpha_G = \sum_M q_M \frac{|G \cap M|}{|M|}. \quad (4)$$

**Interpretation:** Reach treats all followers equally. Attention share recognizes that a voter who follows one outlet is easier to influence through that outlet than a voter who follows many competing outlets.

### 3.3 Candidate quality and media signals

There are two candidates,  $A$  and  $B$ . The relative quality of candidate  $B$  is denoted  $\sigma$ , distributed symmetrically around  $1/2$ . Voters prefer  $B$  when

$$\sigma \geq \frac{1}{2}. \quad (5)$$

Media sources do not directly observe  $\sigma$ . Instead, each source receives  $N$  binary news items. For source  $m$ ,

$$x_m = (x_{m1}, \dots, x_{mN}), \quad \Pr(x_{mi} = 1) = \sigma. \quad (6)$$

A value of one is interpreted as news favorable to candidate  $B$ .

**Interpretation:** The model makes all outlets equally informed in the unbiased benchmark. Political power comes from selection and manipulation of reports, not from one outlet having a better signal technology.

### 3.4 Voter bandwidth

Voters may have limited bandwidth. Even if a voter follows many sources and each source reports many items, the voter observes or remembers only

$$K \in \{1, \dots, N\} \quad (7)$$

randomly selected news items.

**Interpretation:** Bandwidth is central because a biased owner has more influence when voters remember only a few items. Low bandwidth creates random mistakes, and those mistakes can be exploited by biased reporting.

### 3.5 U.S. implementation data

The empirical application uses the 2012 Pew Media Consumption Survey.

- The survey asks which news sources respondents follow and how frequently they follow them.
- The baseline counts sources used daily or almost daily.
- The survey covers television, radio, printed newspapers, websites, and social media.
- The paper combines traditional and website access to the same source, for example print and online access to the *New York Times*.
- Corporate ownership is mapped to media conglomerates such as News Corporation, Comcast, and Time Warner.

**Interpretation:** The key limitation is that the survey does not perfectly measure time spent with each source. The paper therefore uses equal attention across the sources followed by a respondent.

## 4 Theory and empirical specifications

### 4.1 Benchmark with unbiased media

With unbiased media, all outlets report all signals. Let  $s_i$  denote the average of the  $K$  news items observed by voter  $i$ . The voter votes for  $B$  if

$$s_i \geq \frac{1}{2}. \quad (8)$$

**Proposition 1.** With unbiased media, as  $N \rightarrow \infty$ , candidate  $B$  is elected if and only if  $\sigma \geq 1/2$ .

**Interpretation:** Limited bandwidth affects the margin of victory but not the identity of the winner under unbiased reporting. Voters may make mistakes individually, but the electorate chooses the correct candidate in aggregate.

## 4.2 Biased media owner

Consider an owner who controls a subset  $G \subseteq \mathcal{M}$  and wants candidate  $A$  to win regardless of candidate quality. In the worst case, she can suppress or select news items to favor  $A$ .

For a voter segment  $M$ , define the controlled attention share within that segment as

$$g_M = \frac{|G \cap M|}{|M|}. \quad (9)$$

If controlled outlets always report news favorable to  $A$ , then the probability that any remembered item is favorable to  $B$  is

$$(1 - g_M)\sigma. \quad (10)$$

**Interpretation:** The same media owner is more influential in a segment where her outlets represent a larger share of the voter's media bundle.

## 4.3 Vote share under biased reporting

Let  $p_A(g_M, K, \sigma)$  denote candidate  $A$ 's vote share in segment  $M$ . Under manipulation, it is the probability that a voter observes enough zero signals to vote for  $A$ :

$$p_A(g_M, K, \sigma) = \Pr \left( \text{number of remembered favorable-to-}B \text{ items} < \frac{K}{2} \right). \quad (11)$$

Equivalently, it is a binomial cumulative distribution function with success probability  $(1 - g_M)\sigma$ .

**Interpretation:** Manipulation shifts the vote-share function to the right: candidate  $A$  can win even when candidate  $B$  is objectively better.

## 4.4 Media power index

Let  $\sigma^*(K)$  be the highest value of  $\sigma$  for which the biased owner can still get candidate  $A$  elected. The media power index is

$$P_G(K) = 2\sigma^*(K) - 1. \quad (12)$$

**Interpretation:** The index is measured on a quality scale. If  $P_G = 0.10$ , the owner can induce the electorate to choose  $A$  even when  $B$  is better by up to 10 percent of the full quality range.

## 4.5 Upper-bound result

**Proposition 2.** The upper bound to media power is reached when the biased owner selects only news items favorable to candidate  $A$ .

**Proposition 3.** If the share of voters who follow at least one source owned by  $G$  is no greater than 50 percent, then the power index is bounded by

$$P_G(1) = \min \left\{ 1, \frac{\alpha_G}{1 - \alpha_G} \right\}. \quad (13)$$

**Interpretation:** This is the main practical formula. It transforms the attention share of a media owner into a worst-case political influence bound.

## 4.6 Calibration specification

The calibration introduces a sophistication parameter  $\beta$  that scales down the theoretical worst case. In simplified form,

$$P_G^{calibrated} = P_G \times \text{effective naivete among exposed voters.} \quad (14)$$

The paper uses empirical estimates of Fox News effects to infer the sophistication of nonreader television viewers, and then applies the resulting calibration to other organizations.

**Interpretation:** The calibration is not meant to replace the upper-bound index. It gives a lower, empirically disciplined benchmark using existing persuasion estimates.

## 5 Identification, interpretation, and policy design

### 5.1 What the index identifies

The media power index identifies an upper bound on potential political influence, not the actual persuasion effect in a particular election.

- It does not require observing biased reporting.
- It does not require estimating voters' true beliefs.
- It does not require knowing exact attention bandwidth.
- It uses consumption patterns to bound how influential a media owner could be.

**Interpretation:** This makes the index useful for regulation, where policymakers often care about worst-case risks before harm is observed.

### 5.2 Why standard concentration measures are insufficient

Standard market-share measures typically focus on product markets or advertising markets. The media power index differs because:

- cross-platform sources can influence the same voter;
- small economic outlets can be politically influential if they have captive audiences;
- overlapping audiences reduce incremental power;
- power depends on attention, not just reach.

**Interpretation:** A merger that looks harmless within a platform can be politically important if it gives one owner a large share of attention among many voters.

### 5.3 Assumptions behind the upper bound

The theoretical index is deliberately conservative. It assumes:

- the owner is purely partisan;
- the owner can choose reports strategically;
- other outlets are unbiased;
- voters do not change their source choices in response to perceived bias;
- voters may have limited bandwidth.

**Interpretation:** These assumptions make the number a worst-case measure. The calibrated version partially relaxes the naive assumption by using external persuasion estimates.

## 5.4 Incremental policy application

The paper shows how the index can guide merger policy. Regulators can compute the attention share and power index of a merged entity and compare it with alternative buyers or merger partners.

**Interpretation:** The regulatory question becomes: does a merger create an owner with unusually high control over voter attention, even if it does not create a large increase in platform-specific market share?

## 6 Tables and figures

### 6.1 Table 1 and Figure 1: media consumption and unbiased voting

Read:

- Table 1 presents the conceptual media-consumption matrix with nine voter segments and seven sources.
- It distinguishes reach, attention share, and the resulting power index.
- Figure 1 shows candidate  $A$ 's vote share under unbiased reporting for different bandwidth levels  $K$ .
- All curves intersect at  $(1/2, 1/2)$ , so unbiased media elect the better candidate in aggregate.

**Economic message:** Attention share is the basic input to media power, while Figure 1 shows why bandwidth matters for margins even when media are unbiased.

TABLE 1  
MEDIA CONSUMPTION MATRIX

| Segment   | Share | Tv1  | Tv2   | Np1  | Np2  | Np3  | Web1 | Web2  |
|-----------|-------|------|-------|------|------|------|------|-------|
| 1         | 20%   | ■    |       |      |      |      |      |       |
| 2         | 10%   | ■    |       | ■    |      |      |      |       |
| 3         | 10%   |      |       | ■    |      |      |      |       |
| 4         | 10%   |      | ■     |      |      |      |      | ■     |
| 5         | 10%   |      | ■     |      | ■    |      | ■    |       |
| 6         | 10%   |      | ■     |      | ■    |      | ■    |       |
| 7         | 10%   |      | ■     |      |      | ■    | ■    |       |
| 8         | 10%   |      |       |      |      | ■    | ■    | ■     |
| 9         | 10%   |      |       |      | ■    | ■    | ■    | ■     |
| Reach     |       | 30%  | 40%   | 20%  | 30%  | 30%  | 50%  | 40%   |
| Attention |       | 25%  | 14.1% | 15%  | 8.3% | 9.1% | 15%  | 13.3% |
| $\Pi$     |       | .333 | .164  | .176 | .090 | .101 | .176 | .152  |

NOTE.—Example of a media consumption matrix with seven media sources: two television networks (Tv1, Tv2), three newspapers (Np1, Np2, Np3), and two online sources (Web1, Web2). Voters are divided into nine segments depending on the sources they follow. In each segment voters devote equal attention to each source indicated by a box. The *reach* of a source is the percentage of voters who follow that source. The *attention share* of a source is the weighted average of that source's attention share in each segment. The variable  $\Pi$  is the power index of the correspondent source as defined in proposition 3.

Table 1: media consumption matrix

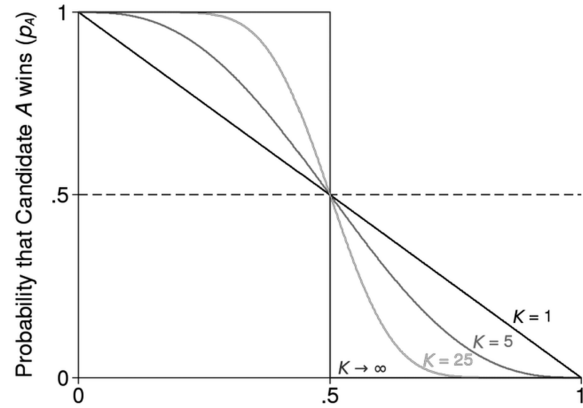


Figure 1: unbiased reporting

### 6.2 Figure 2: biased reporting in the numerical example

Read:

- Figure 2 shows how vote-share curves shift when newspaper 2 is biased in the Table 1 example.
- Segments not reading newspaper 2 retain the unbiased vote-share function.
- Segments that read newspaper 2 have vote-share functions shifted to the right.
- The aggregate curve identifies the highest  $\sigma$  for which the biased owner can still elect candidate  $A$ .

**Economic message:** Biased media power arises because the owner shifts the election cutoff: candidate  $A$  can win even when candidate  $B$  is better.

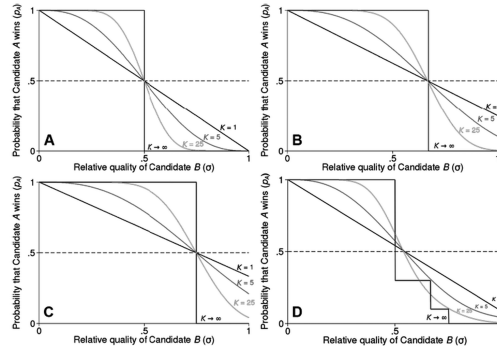


Figure 2: vote-share curves when newspaper 2 is biased

### 6.3 Table 2 and Table 3: U.S. media power and calibration

Read:

- Table 2 reports media power indices for major U.S. media organizations using 2012 Pew data.
- News Corporation has the highest baseline power index, 0.221.
- Comcast is second with 0.153, and Time Warner is third with 0.086.
- Television-based organizations dominate because they have large reach and audiences that follow relatively few other sources.
- Table 3 calibrates the theoretical bounds using estimates of Fox News effects from the empirical media literature.
- Calibrated power values are much smaller than worst-case values, but the relative ranking remains broadly similar.

**Economic message:** The empirical application shows that political media power is concentrated in television-centered organizations, and that this conclusion survives calibration.



TABLE 2  
MEDIA POWER INDICES

| Media Organization      | Reach (1) | Attention Share (2) | 5+ Sources (3) | II (4) | CPI (5) | Registered Voters (6) | Independent Voters (7) | Swing Voters (8) |
|-------------------------|-----------|---------------------|----------------|--------|---------|-----------------------|------------------------|------------------|
| News Corp. (Fox, WSJ)   | .311      | .181                | 21.8           | .221   | 19      | .225                  | .215                   | .228             |
| Comcast (NBC, MSNBC)    | .268      | .133                | 41.8           | .153   | 166     | .161                  | .127                   | .189             |
| Time Warner (CNN, Time) | .209      | .079                | 32.9           | .086   | 24      | .081                  | .082                   | .082             |
| ABC                     | .143      | .062                | 25.2           | .066   | 4       | .066                  | .048                   | .069             |
| NPR                     | .141      | .056                | 22.6           | .059   | 26      | .061                  | .076                   | .061             |
| Yahoo                   | .126      | .054                | 16.4           | .057   | n.a.    | .047                  | .066                   | .042             |
| CBS                     | .087      | .035                | 25.0           | .036   | 1       | .035                  | .034                   | .029             |
| MSN                     | .094      | .032                | 32.1           | .033   | 25      | .031                  | .035                   | .031             |
| NY Times                | .079      | .030                | 41.2           | .031   | 5       | .032                  | .028                   | .010             |
| Viacom                  | .063      | .028                | 48.6           | .028   | 0       | .025                  | .042                   | .024             |
| PBS                     | .079      | .024                | 31.3           | .025   | 8       | .025                  | .024                   | .025             |
| Rush Limbaugh           | .063      | .021                | 25.0           | .022   | n.a.    | .023                  | .014                   | .024             |
| Google                  | .055      | .021                | 22.5           | .022   | 88      | .018                  | .027                   | .017             |
| USA Today               | .047      | .019                | 28.6           | .020   | 5       | .020                  | .017                   | .020             |
| AOL                     | .031      | .013                | 17.8           | .013   | .7      | .014                  | .019                   | .006             |

SOURCE.—Pew Media Consumption Survey, 2012.

NOTE.—Col. 1: Reach is the share of media users who follow that source regularly. Col. 2: Attention share of that media source (average individual attention share, where the attention share of that source for an individual who follows  $N$  sources is  $1/N$  if he follows that source and zero otherwise). Col. 3: Percentage of users of that source who follow at least five sources. Col. 4: Media power index II as in proposition 3, computed on the basis of daily usage for all the United States. Col. 5: Company power index (CPI): the average squared market share across platforms weighted by total revenues of all firms in each platform, computed in 2011 or later years (Noam 2009, 365). The range is 0 (perfect competitor) to 10,000 (monopolist). Col. 6: As in col. 4 but restricted to the sample of respondents who report being registered voters. Col. 7: As in col. 4 but restricted to respondents who self-identify as independent. Col. 8: Same as in col. 4 but restricted to residents of swing states: CO, FL, IA, NE, NH, NC, OH, VA, and WI.

Table 2: media power indices

TABLE 3  
CALIBRATION

| Media Organization                | Baseline II (1) | Total Attention (2) | Nonreaders' Attention (3) | DellaVigna/Kaplan (4) | Martin/Yurukoglu (5) |
|-----------------------------------|-----------------|---------------------|---------------------------|-----------------------|----------------------|
| News Corp. (Fox, WSJ)             | .221            | .181                | .519                      | .018                  | .104                 |
| Comcast (NBC, MSNBC)              | .153            | .133                | .414                      | .010                  | .058                 |
| Time Warner (CNN, Time)           | .086            | .079                | .455                      | .007                  | .037                 |
| ABC                               | .066            | .062                | .552                      | .006                  | .035                 |
| NPR                               | .059            | .056                | .502                      | .005                  | .029                 |
| Yahoo                             | .057            | .054                | .686                      | .007                  | .038                 |
| CBS                               | .036            | .035                | .551                      | .004                  | .019                 |
| MSN                               | .033            | .032                | .603                      | .004                  | .019                 |
| NY Times                          | .031            | .030                | .000                      | .000                  | .000                 |
| Viacom                            | .028            | .028                | .543                      | .003                  | .015                 |
| PBS                               | .025            | .024                | .417                      | .002                  | .010                 |
| Google                            | .022            | .021                | .616                      | .002                  | .013                 |
| Rush Limbaugh                     | .022            | .021                | .561                      | .002                  | .012                 |
| USA Today                         | .020            | .019                | .000                      | .000                  | .000                 |
| AOL                               | .013            | .013                | .556                      | .001                  | .007                 |
| Share of sophisticated readers    | 0%              |                     |                           | 100%                  | 100%                 |
| Share of sophisticated nonreaders | 0%              |                     |                           | 76%                   | 0%                   |

SOURCE.—Pew Media Consumption Survey, 2012.

NOTE.—Col. 1: Media power index II as in proposition 3 (same as col. 4 in table 2). Col. 2: Attention share of media organizations (same as col. 2 in table 2). Col. 3: Attention share of media organization excluding readers. Col. 4: Calibrated media power index based on estimates by DellaVigna and Kaplan (2007) for broadcast media and Chiang and Knight (2011) and Gentzkow et al. (2011) for newspaper readers. Col. 5: Calibrated media power index on the basis of Martin and Yurukoglu (2016) for broadcast media and Chiang and Knight (2011) and Gentzkow et al. (2011) for newspaper readers.

Table 3: calibrated media power

## 6.4 Table 4 and Table 5: merger analysis

Read:

- Table 4 revisits the numerical example and compares two possible acquisitions of newspaper 3.
- A reach-based approach says the television acquisition is more dangerous, but the media power index says the newspaper acquisition is more powerful.
- Table 5 considers hypothetical pairwise mergers among major U.S. media organizations.
- News Corporation plus Time Warner has the largest hypothetical merged power index.

**Economic message:** Merger risk depends on attention overlap, not just reach. The same target can create different political power depending on the buyer's existing audience portfolio.

TABLE 4  
MERGER ANALYSIS: NUMERICAL EXAMPLE

|           | MERGING SOURCES |      |      | MERGED ENTITIES |         |
|-----------|-----------------|------|------|-----------------|---------|
|           | Tv2             | Np1  | Np3  | Tv2+Np3         | Np1+Np3 |
| Reach     | 40%             | 20%  | 30%  | 60%             | 50%     |
| Attention | 14.1%           | 15%  | 9.1% | 23.2%           | 24.1%   |
| II        | .165            | .176 | .100 | .304            | .318    |

NOTE.—Effect of the acquisition of newspaper 3 (Np3) by either television network 2 (Tv2) or newspaper 1 (Np1), based on the media consumption matrix in table 1. Reach is the percentage of voters following that source or set of sources. Attention is the share of that source or set of sources. The variable II is the power index of that source or set of sources (as defined in proposition 3).

Table 4: numerical merger example

TABLE 5  
MERGER ANALYSIS: HYPOTHETICAL MERGERS OF EXISTING ORGANIZATIONS

|           | MERGING SOURCES |            |             |        | MERGED ENTITIES |       |       |            |
|-----------|-----------------|------------|-------------|--------|-----------------|-------|-------|------------|
|           | CBS             | News Corp. | Time Warner | Viacom | News Corp.      | CBS   | CBS   | News Corp. |
| Reach     | 8.7%            | 31.1%      | 20.9%       | 6.3%   | 36.8%           | 26.4% | 14.3% | 43.8%      |
| Attention | 3.5%            | 18.1%      | 7.9%        | 2.8%   | 21.6%           | 11.4% | 6.2%  | 28.0%      |
| II        | .036            | .221       | .086        | .028   | .275            | .128  | .066  | .352       |

NOTE.—Hypothetical pairwise mergers between CBS, News Corp., Time Warner, and Viacom, based on 2012 Pew data, as reported in table 2. Reach is the percentage of voters following that source or set of sources. Attention is the share of that source or set of sources. The variable II is the power index of that source or set of sources (as defined in proposition 3).

Table 5: hypothetical mergers of existing organizations

## 6.5 Table B1: robustness and alternative specifications

Read:

- Table B1 reports alternative media power indices using 2010 data, weekly usage, an extreme worst case, respondents under 40, and New York State respondents.
- News Corporation remains among the most powerful organizations across specifications.
- The extreme worst case raises all power indices, especially for large television organizations.

- Younger respondents show somewhat different rankings, with online organizations gaining relative strength.

**Economic message:** The broad pattern is robust: traditional television-centered organizations remain powerful, but demographics and usage definitions change the exact values.

TABLE B1  
MEDIA POWER INDICES

| Media Organization      | Baseline II<br>(1) | 2010<br>(2) | Weekly<br>Usage<br>(3) | Extreme<br>Worst Case<br>(4) | Younger<br>than 40<br>(5) | New York<br>State<br>(6) |
|-------------------------|--------------------|-------------|------------------------|------------------------------|---------------------------|--------------------------|
| News Corp. (Fox, WSJ)   | .221               | .228        | .244                   | .419                         | .174                      | .120                     |
| Comcast (NBC, MSNBC)    | .153               | .144        | .196                   | .333                         | .103                      | .159                     |
| Time Warner (CNN, Time) | .086               | .096        | .107                   | .240                         | .107                      | .134                     |
| ABC                     | .066               | .063        | .073                   | .170                         | .025                      | .049                     |
| NPR                     | .059               | .049        | .058                   | .170                         | .084                      | .028                     |
| Yahoo                   | .057               | .067        | .021                   | .162                         | .108                      | .061                     |
| CBS                     | .036               | .035        | .045                   | .111                         | .017                      | .055                     |
| MSN                     | .033               | .043        | .014                   | .122                         | .056                      | .031                     |
| NY Times                | .031               | .026        | .032                   | .101                         | .052                      | .067                     |
| Viacom                  | .028               | .034        | .069                   | .081                         | .077                      | .032                     |
| PBS                     | .025               | .017        | .043                   | .102                         | .007                      | .023                     |
| Rush Limbaugh           | .022               | .016        | .031                   | .082                         | .022                      | .004                     |
| Google                  | .022               | .023        | .008                   | .071                         | .047                      | .036                     |
| USA Today               | .020               | .015        | .042                   | .060                         | .024                      | .031                     |
| AOL                     | .013               | .020        | .004                   | .039                         | .012                      | .031                     |

SOURCE.—Pew Media Consumption Survey. All data are for the year 2012 except where noted.

NOTE.—Col 1: Media power index II as in proposition 3, computed on the basis of daily usage for all the United States. Col. 2: As in col. 1 but computed for 2010. Col. 3: As in col. 1 but computed on the basis of weekly usage. Col. 4: As in col. 1 but using the extreme worst-case media power index computed under the assumption that bandwidth can differ across segments. Col. 5: As in col. 1 but restricted to respondents under 40 years of age. Col. 6: As in col. 1 but restricted to residents of New York State.

Table B1: additional media power specifications

## 7 Short summary

- The paper defines media power as the ability to induce voters to make electoral decisions they would not make under unbiased reporting.
- The analysis begins from individual-level media bundles rather than media platforms or advertising markets.
- Attention share is the key empirical object: it discounts reach by the number of other sources followed by the same voters.
- The model has two candidates, noisy media signals, voters with possible bandwidth limits, and a potentially biased media owner.
- Under unbiased media, the better candidate wins in aggregate as the number of signals grows large.
- Under biased reporting, an owner can shift vote-share functions by selectively reporting favorable news.
- The media power index is  $P_G(K) = 2\sigma^*(K) - 1$ .
- Under a 50 percent reach condition, the worst-case power bound is  $\min\{1, \alpha_G/(1 - \alpha_G)\}$ .
- Applying the index to 2012 Pew data, News Corporation is the most powerful organization in the baseline specification.
- Calibration using Fox News persuasion estimates lowers absolute values but preserves many rankings.
- Merger analysis shows why reach and platform rules can miss politically relevant concentration.

## 8 Conclusion

### 8.1 Core conclusion

Prat develops a voter-centered index of media power. The index measures the maximum electoral distortion that a media owner could generate under worst-case reporting and attention assumptions. Its empirical application shows that U.S. media power is concentrated among television-centered organizations, especially News Corporation.

### 8.2 Economic intuition

A media owner is powerful when many voters rely heavily on that owner's sources and have limited exposure to alternative information. Reach alone is insufficient: what matters is how much of each voter's attention the owner controls. Biased reporting matters most when voters recall only a small number of news items and when the owner accounts for a large share of those items.

### 8.3 Incremental takeaway

The paper turns media-concentration policy into a problem of information aggregation. It provides a tractable bridge between political economy theory and merger analysis: start from individual media consumption, compute attention overlap, and evaluate how much electoral influence an owner could exert in the worst case.

### 8.4 Policy takeaway

Regulators should not evaluate media mergers solely through platform-specific market shares. They should also ask whether the transaction gives one owner excessive control over the information environment of individual voters.